SQL Game: mystery.knightlab.com

> YouTube Streaming: https://youtube.com/live/A8-T6vuDJ_w?feature=share

General Information

> that the crime was a **murder**.

> occurred sometime on **Jan.15, 2018**

> took place in **SQL City**.

Table List

```
SELECT name
  FROM sqlite_master
 where type = 'table'
```

name
crime_scene_report
drivers_license
facebook_event_checkin
interview
get_fit_now_member
get_fit_now_check_in
solution
income
person

Table Structure: `crime_scene_report`

```
SELECT sql
  FROM sqlite_master
 where name = 'crime_scene_report'
```

CREATE TABLE crime_scene_report (date integer, type text, description text, city text)

`crime_scene_report`: Type, date, city

```
select * from crime_scene_report
where type='murder'
and date='20180115'
and city like 'sql%'
```

date	type	description	city
20180115	murder	Security footage shows that there were 2 witnesses. The first witness lives at the last house on "Northwestern Dr". The second witness, named Annabel, lives somewhere on "Franklin Ave".	SQL City

> "... there were **2 witnesses**."

1. > "... **last house** on **Northwestern Dr**".
2. > "... **Annabel** on **Franklin Ave**".

Find the 2 witnesses

`person`: 1st Witness

```
select * from person
where address_street_name='Northwestern Dr'
and address_number > 4900
```

id	name	license_id	address_number	address_street_name	ssn
14887	Morty Schapiro	118009	4919	Northwestern Dr	111564949

`person`: 2nd Witness

```
select * from person
where name like '%Annabel%'
and address_street_name like 'frank%'
```

id	name	license_id	address_number	address_street_name	ssn
16371	Annabel Miller	490173	103	Franklin Ave	318771143

> 1st witness is **Morty Schapiro** with licnse_id **118009**, 2nd witness is **Annabel Miller** with license_id **490173**.

Find the information based on `interview`

```
select * from interview
where person_id='14887'

union

select * from interview
where person_id='16371'
```

person_id	transcript
14887	I heard a gunshot and then saw a man run out. He had a "Get Fit Now Gym" bag. The membership number on the bag started with "48Z". Only gold members have those bags. The man got into a car with a plate that included "H42W".
16371	I saw the murder happen, and I recognized the killer from my gym when I was working out last week on January the 9th.

Gym membership based on 2nd witness

```
select * from get_fit_now_member
where person_id='16371'
```

id	person_id	name	membership_start_date	membership_status
90081	16371	Annabel Miller	20160208	gold

```
select * from get_fit_now_check_in
where membership_id='90081'
```

membership_id	check_in_date	check_in_time	check_out_time
90081	20180109	1600	1700

> Since the 2nd witness mentioned abot the time explicitly "... when I was working out", therefore we can filter the time.

```
select * from get_fit_now_check_in
where check_in_date='20180109'
and check_out_time > '1600'
```

membership_id	check_in_date	check_in_time	check_out_time
48Z7A	20180109	1600	1730
48Z55	20180109	1530	1700
90081	20180109	1600	1700

> Suspect will be 2 people with `membership_id`: **48Z7A** and **48Z55**.

`driver_license` and Gym bag based on 1st witness

```
select * from drivers_license
where plate_number like '%H42W%'
```

id	age	height	eye_color	hair_color	gender	plate_number	car_make	car_model
183779	21	65	blue	blonde	female	H42W0X	Toyota	Prius
423327	30	70	brown	brown	male	0H42W2	Chevrolet	Spark LS
664760	21	71	black	black	male	4H42WR	Nissan	Altima

```
select * From get_fit_now_member
where id like '48Z%'
```

id	person_id	name	membership_start_date	membership_status
48Z38	49550	Tomas Baisley	20170203	silver
48Z7A	28819	Joe Germuska	20160305	gold
48Z55	67318	Jeremy Bowers	20160101	gold

2 Suspects.

```
select * From get_fit_now_member
where id like '48Z%'
and id='48Z7A'

union

select * from get_fit_now_member
where id like '48Z%'
and id='48Z55'
```

id	person_id	name	membership_start_date	membership_status
48Z55	67318	Jeremy Bowers	20160101	gold
48Z7A	28819	Joe Germuska	20160305	gold

The suspect

```
select * from person
where license_id='183779'
union
select * from person
where license_id='423327'
union
select*from person
where license_id='664760'
```

id	name	license_id	address_number	address_street_name	ssn
51739	Tushar Chandra	664760	312	Phi St	137882671
67318	Jeremy Bowers	423327	530	Washington Pl, Apt 3A	871539279
78193	Maxine Whitely	183779	110	Fisk Rd	137882671

Since the `Jeremy Bowers` name appeared on both evidence, then we can surely conclude that he is **the one who commited murder crime**, therefore **CASE CLOSED**.

Result

```
INSERT INTO solution VALUES (1, 'Jeremy Bowers');
    SELECT value FROM solution;
```

> Congrats, you found the murderer! But wait, there's more... If you think you're up for a challenge, try querying the interview transcript of the murderer to find the real villain behind this crime. If you feel especially confident in your SQL skills, try to complete this final step with no more than 2 queries.
> Use this same INSERT statement with your new suspect to check your answer.

Deeper research for the **Real Villain**.

`interview` transcription from murderer

```
select * from interview
where person_id='67318'
```

person_id	transcript
67318	I was hired by a woman with a lot of money. I don't know her name but I know she's around 5'5" (65") or 5'7" (67"). She has red hair and she drives a Tesla Model S. I know that she attended the SQL Symphony Concert 3 times in December 2017.

In the transcript above, we can list down the information of the **real villain**.

1. woman
2. a lot of money
3. 5'5" (65") or 5'7" (67")
4. Red hair
5. Tesla Model S
6. Attended the **SQL Symphony Concert 3 times** in **December 2017**

Information based on `drivers_license`

- [x] woman
- [x] red hair color
- [x] around 65" or 67" height
- [x] Tesla Model S car

```
select*from drivers_license
where hair_color='red'
and gender='female'
and height between 65 and 67
and car_make like 'tesla'
```

id	age	height	eye_color	hair_color	gender	plate_number	car_make	car_model
202298	68	66	green	red	female	500123	Tesla	Model S
291182	65	66	blue	red	female	08CM64	Tesla	Model S
918773	48	65	black	red	female	917UU3	Tesla	Model S

Information based on `facebook_event_checkin`

- [x] Attended the SQL Symphony Concert 3 times in December 2017

```
select person_id, count(*)
from facebook_event_checkin
where date like '201712%'
and event_name like 'sql%'
group by person_id
having count(*)=3
```

person_id	count(*)
24556	3
99716	3

Join all the information

> From the `person` we have

```
select * from person
where license_id='202298'

union
select * from person
where license_id='291182'

union
select * from person
where license_id='918773'
```

id	name	license_id	address_number	address_street_name	ssn
78881	Red Korb	918773	107	Camerata Dr	961388910
90700	Regina George	291182	332	Maple Ave	337169072
99716	Miranda Priestly	202298	1883	Golden Ave	987756388

> Since we already have `person_id` then we can bravely to make `Miranda Priestly` as the **real villain** behind this murder crime.

Or... we can simply use this query

```
select * from person
where id in ('24556','99716')
and license_id in ('202298','291182','918773')
```

id	name	license_id	address_number	address_street_name	ssn
99716	Miranda Priestly	202298	1883	Golden Ave	987756388

Result

```
INSERT INTO solution VALUES (1, 'Miranda Priestly');
    SELECT value FROM solution;
```

> Congrats, you found the brains behind the murder! Everyone in SQL City hails you as the greatest SQL detective of all time. Time to break out the champagne!